

AHMAD RAZA

FULL STACK AI ENGINEER | AI AUTOMATION ENGINEER | BACKEND ENGINEER

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PROFESSIONAL SUMMARY

Full Stack AI and Automation Engineer with 1+ years of hands-on project experience building RAG systems, AI agents, workflow automations, and API-driven web applications. Strong in Python, FastAPI, PostgreSQL/pgvector, n8n, LangChain/LangGraph, Docker, LLM integrations, secure authentication, and REST API architecture. Built a production-style AI Engineering Copilot with semantic/MMR retrieval, source-grounded responses, structured AI workflows, evaluation, automated testing, and containerized deployment. Based in Lahore and ready to contribute to commercially deployed AI products in a fast-moving startup environment.

TECHNICAL SKILLS

AI Engineering & RAG: Retrieval-Augmented Generation, RAG pipelines, AI agents, LLM integrations, LangChain, LangGraph, Langsmith, semantic search, MMR, embeddings, vector databases, document ingestion/parsing, prompt engineering, structured outputs, guardrails, RAG evaluation, Ollama, Sentence Transformers

Full-Stack & Backend: Python, FastAPI, Node.js, Express.js, React.js, REST APIs, Pydantic, SQLAlchemy, HTTPX, CRUD, JWT, authentication, authorization, role-based access control, API integrations, Postman

Automation & Voice AI: n8n, webhooks, triggers, API orchestration, conditional routing, data transformation, WhatsApp chatbots, ElevenLabs voice agents, business process automation, notifications, error handling, workflow debugging

Data, DevOps & Quality: PostgreSQL, pgvector, SQL, SQLite, database design, Docker, Docker Compose, Streamlit, Git/GitHub, Pytest, Ruff, C, C++

SELECTED PROJECTS

AI Engineering Copilot | Sole Developer & AI Engineer | Python, FastAPI, Streamlit, PostgreSQL, pgvector, Ollama, Docker

- Architected and built a full-stack RAG platform that safely ingests ZIP codebases, discovers and parses source/documentation files, creates code-aware chunks and embeddings, and stores searchable vectors in PostgreSQL with pgvector.
- Implemented semantic and MMR retrieval, source-grounded Q&A with file/line citations, confidence calibration, missing-context detection, retrieval diagnostics, prompt-injection-resistant context handling, and Pydantic-validated outputs.
- Delivered AI modules for code review, feature planning, documentation generation, and retrieval/RAG evaluation; containerized Streamlit, FastAPI, and PostgreSQL with Docker Compose and added API, schema, retrieval, validation, and smoke tests.

AI Automation & Voice Agent Workflows | Automation Engineer | n8n, WhatsApp, Webhooks, APIs, ElevenLabs

- Built multi-step business automations for real-estate lead capture, university TA/student support and notifications, and customer-service workflows using n8n, webhooks, external APIs, conditional routing, data transformation, and error handling.
- Developed WhatsApp chatbot and ElevenLabs voice-agent workflows that collected user data, routed requests, triggered follow-up actions, and automated notifications.

FastCart API | Backend Developer | FastAPI, Python, SQL, Pydantic, Authentication, REST APIs

- Built an e-commerce REST API for products, carts, cart items, and users with Pydantic schemas, SQL persistence, CRUD operations, authentication/authorization, validation, error handling, and Postman testing.

SmartRecipe Hub | Full-Stack Developer | React.js, Node.js, Express, SQLite, JWT/bcrypt

- Built a full-stack recipe and meal-planning application with secure authentication, recipe search/filtering, pantry tracking, weekly planning, shopping lists, ratings/reviews, and role-based admin access.

EDUCATION & ACHIEVEMENT

BS Software Engineering | Sep 2024 - Present

Punjab University College of Information Technology, University of the Punjab | Lahore, Pakistan

1st Place - Olympiad 5.0 Project Exhibition, University of Lahore | Braille-to-Text Converter

- Won 1st place for a low-cost microcontroller-based accessibility solution using push-button Braille input, LEDs, and mobile text output.